

SECTION A

1 **(short)** Find the solutions of $z^{2i} + 2 = 3z^i$ in Cartesian form. [10]

2 **(short)** Find $\lim_{y \rightarrow \infty} y \log(1 + \frac{x}{y})$. [10]

3 **(short)** Find all solutions of the difference equation $x_{n+2} + x_{n+1} - 6x_n = 0$, subject to $x_0 = x_1$. [10]

4 **(long)** In three-dimensional space $\mathbf{x}$ satisfies

$$4\mathbf{x} \cdot \mathbf{m} = \mathbf{x} \cdot \mathbf{x} - (\mathbf{x} \cdot \mathbf{m})^2 - (\mathbf{x} \cdot \mathbf{n})^2, \quad (1)$$

where $\mathbf{m}$ and $\mathbf{n}$ are two orthogonal unit vectors.

(a) Explain why it must be possible to choose α, β, γ such that

$$\mathbf{x} = \alpha\mathbf{m} + \beta\mathbf{n} + \gamma\mathbf{m} \times \mathbf{n}. \quad [5]$$

(b) Show that if $\mathbf{x}$ satisfies eq. (1) then

$$4\alpha = \gamma^2$$

and give the range of values for β . Provide a geometrical interpretation for eq. (1). [10]

(c) For any $\mathbf{x}$ satisfying eq. (1) obtain expressions for the shortest distance between:

(i) $\mathbf{x}$ and the plane $\mathbf{r} \cdot \mathbf{m} = -1$;

(ii) $\mathbf{x}$ and the line $\mathbf{r} = \mathbf{m} + \lambda\mathbf{n}$.

How are these distances related? [15]